The Fundamentals of Statistical Inference in the Simple Linear Regression Model

1. Introduction to Statistical Inference

1.1 Starting Point

We have derived **point estimators** of all the unknown population parameters in the Classical Linear Regression Model (CLRM) for which the **population regression equation, or PRE**, is

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad (i = 1, \ldots, N) \qquad \textbf{(1)}$$

- The **unknown parameters** of the PRE are

  (1) the regression coefficients β_0 and β_1

  and

  (2) the error variance σ^2.

Statistical Inference in the Simple Linear Regression Model

- The ***point*** **estimators** of these unknown population parameters are

  (1) the ***unbiased*** **OLS regression coefficient estimators** $\hat{\beta}_0$ **and** $\hat{\beta}_1$

  and

  (2) the ***unbiased*** **error variance estimator** $\hat{\sigma}^2$ given by the formula

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{(N-2)} = \frac{RSS}{(N-2)} \quad \text{where} \quad \hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \quad (i = 1, \ldots, N).$$

Note: $\hat{\sigma}^2$ **is an** ***unbiased*** **estimator of the error variance** σ^2:

$$E(\hat{\sigma}^2) = \sigma^2 \quad \text{because} \quad E(RSS) = E\left(\Sigma_i \hat{u}_i^2\right) = (N-2)\sigma^2.$$

1.2 Nature of Statistical Inference

Statistical inference consists essentially of **using the point estimates** $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\sigma}^2$ of the unknown population parameters β_0, β_1 and σ^2 **to make statements about the true values of** β_0 **and** β_1 ***within specified margins of statistical error***.

1.3 Two Related Approaches to Statistical Inference

1. **Interval estimation (or the confidence-interval approach)** involves using the point estimates $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\sigma}^2$ to construct confidence intervals for the regression coefficients that contain the true population parameters β_0 and β_1 ***with some specified probability***.

1.2 Nature of Statistical Inference

2. **<u>Hypothesis testing</u> (or the test-of-significance approach)** involves using the point estimates $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\sigma}^2$ to test hypotheses or assertions about the true population values of β_0 and β_1.

- These two approaches to statistical inference are mutually complementary and equivalent in the sense that **they yield identical inferences** about the true values of the population parameters.

- For both types of statistical inference, we need to know the ***form* of the sampling distributions** (or **probability distributions**) of the OLS coefficient estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ and the unbiased (degrees-of-freedom-adjusted) error variance estimator $\hat{\sigma}^2$.

  This is the purpose of the **error normality assumption A9**.

1.4 The Objective: Feasible Test Statistics for $\hat{\beta}_0$ and $\hat{\beta}_1$

For both forms of statistical inference -- interval estimation and hypothesis testing -- it is necessary to obtain ***feasible*** **test statistics for each of the OLS coefficient estimators** $\hat{\beta}_0$ **and** $\hat{\beta}_1$ in the **OLS sample regression equation (OLS-SRE)**

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i \qquad (i = 1,\ldots,N) \qquad \textbf{(2)}$$

1. The **OLS slope coefficient estimator** $\hat{\beta}_1$ can be written in deviation-from-means form as:

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2} \qquad \textbf{(2.1)}$$

where $x_i \equiv X_i - \bar{X}$, $y_i \equiv Y_i - \bar{Y}$, $\bar{X} = \sum_i X_i / N$, and $\bar{Y} = \sum_i Y_i / N$.

2. The **OLS intercept coefficient estimator** $\hat{\beta}_0$ can be written as:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

What's a Feasible Test Statistic?

A ***feasible test statistic*** is **a sample statistic** that satisfies **two properties**:

1. It **has a known probability distribution** for the true population value of some parameter(s).

2. It **is a function only of sample data --** i.e., its value can be computed for some hypothesized value of the unknown parameter(s) using only sample data.

The **formula** for a feasible test statistic **must contain no unknown parameters** other than the parameter(s) of interest, i.e., the parameters being tested.

2. The Normality Assumption A9

2.1 Statement of Normality Assumption

The normality assumption states that the random error terms u_i in the population regression equation (PRE)

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad (i = 1, \ldots, N) \qquad (3)$$

are ***independently* and *identically* distributed as the *normal* distribution** with

(1) *zero* means: $E(u_i | X_i) = 0 \quad \forall \; i;$

(2) *constant* variances: $Var(u_i | X_i) = E(u_i^2 | X_i) = \sigma^2 > 0 \quad \forall \; i; \quad$ **... (A9)**

(3) *zero* covariances: $Cov(u_i, u_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0 \;\forall\; i \neq s.$

Compact Forms of the Normality Assumption A9

(1) $\mathbf{u_i \sim N(0, \sigma^2) \quad \forall\ i = 1, \ldots, N}$ **... (A9.1)**

where $N(0, \sigma^2)$ denotes a normal distribution with mean or expectation equal to 0 and variance equal to σ^2 and the symbol "~" means "is distributed as".

NOTE: The normal distribution has only two parameters: its mean or expectation, and its variance. The probability density function, or pdf, of the normal distribution is defined as follows: if a random variable $X \sim N(\mu, \sigma^2)$, then the pdf of X is denoted as $f(X_i)$ and takes the form

$$f(X_i) = (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left[-\frac{(X_i - \mu)^2}{2\sigma^2}\right]$$

where X_i denotes a *realized or observed value* of X.

Compact Forms of the Normality Assumption A9

(2) $\quad$ **u_i are iid as $N(0, \sigma^2)$** $\quad$ **... (A9.2)**

where "iid" means "independently and identically distributed".

- Parts (1) and (2) of A9 -- the assumptions of zero means and constant variances -- are the "identically" part of the iid specification.

- Part (3) of A9 -- the assumption of zero error covariances -- is the "independently" part of the iid specification.

(3) $\quad$ **u_i are NID$(0, \sigma^2)$** $\quad$ **... (A9.3)**

where NID$(0, \sigma^2)$ means "normally and independently distributed with zero mean and constant variance σ^2".

2.2 Implications of Normality Assumption A9 for the Distribution of Y_i

The error normality assumption A9 implies that the **sample values Y_i** of the regressand (or dependent variable) Y **are also *normally distributed***.

- **<u>Linearity Property of Normal Distribution</u>:** Any linear function of a normally distributed random variable is itself normally distributed.

- The **PRE**, or **population regression equation**, states that **Y_i is a *linear* function of the error terms u_i**:

$$Y_i = \beta_0 + \beta_1 X_i + u_i \qquad (4)$$

- Since **any *linear* function of a *normally distributed random variable* is itself normally distributed**, the error normality assumption A9 implies that

$$\mathbf{Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, \ldots, N \qquad (5.1)}$$

2.2 Implications of Normality Assumption A9 for the Distribution of Y_i

$$\mathbf{Y_i \text{ are NID}(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, \ldots, N} \qquad \mathbf{(5.2)}$$

That is, the **Y_i are *normally* and *independently distributed* (NID)** with

(1) conditional means

$$E(Y_i | X_i) = \beta_0 + \beta_1 X_i \qquad \forall\, i = 1, \ldots, N;$$

(2) conditional variances

$$Var(Y_i | X_i) = E(u_i^2 | X_i) = \sigma^2 \qquad \forall\, i = 1, \ldots, N;$$

(3) conditional covariances

$$Cov(Y_i, Y_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0 \qquad \forall\, i \neq s.$$

3. The Sampling Distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$

To obtain the sampling distributions of the OLS coefficient estimators $\hat{\beta}_0$ and $\hat{\beta}_1$, we make use of

1. the **linearity property** of the **OLS coefficient estimators $\hat{\beta}_0$ and $\hat{\beta}_1$**;

and

2. the **normality assumption A9**.

3.1 Linearity Property of the OLS Coefficient Estimators $\hat{\beta}_0$ and $\hat{\beta}_1$

The linearity property of the OLS coefficient estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ means that the normality of u_i and Y_i carries over to the sampling distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$. For example, recall that the OLS slope coefficient estimator $\hat{\beta}_1$ can be written as the following linear functions of the Y_i and u_i:

$$\hat{\beta}_1 = \Sigma_i k_i Y_i$$

$$= \beta_1 + \Sigma_i k_i u_i \qquad (6)$$

where the observation weights, the k_i, are defined as

$$k_i = \frac{x_i}{\Sigma_i x_i^2} \qquad (i = 1, \ldots, N).$$

- ***Result:*** Equation (6) indicates that the **OLS slope coefficient estimator** $\hat{\beta}_1$ is a ***linear* function** of both the ***observed*** **Y_i values** and the ***unobserved*** **u_i values**.

3.2 Implications of Normality Assumption and Linearity Property

Because any linear function of normally distributed random variables is itself normally distributed, the sampling distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$ are also normally distributed. The reasoning is as follows:

$$\text{normality of } u_i \Rightarrow \text{normality of } Y_i \Rightarrow \text{normality of } \hat{\beta}_0 \text{ and } \hat{\beta}_1.$$

There are **<u>four</u> specific distributional implications of the error normality assumption A9**.

<u>Implication 1</u>: The **sampling distribution of the OLS slope coefficient estimator** $\hat{\beta}_1$ **is *normal*** with mean $E(\hat{\beta}_1) = \beta_1$ and variance $Var(\hat{\beta}_1) = \sigma^2 / \sum_i x_i^2$: i.e.,

$$\hat{\beta}_1 \sim N\left(\beta_1, Var(\hat{\beta}_1)\right) = N\left(\beta_1, \frac{\sigma^2}{\sum_i x_i^2}\right).$$

The normality of the sampling distribution of $\hat{\beta}_1$ implies that the **Z-statistic** $Z(\hat{\beta}_1) = (\hat{\beta}_1 - \beta_1)/se(\hat{\beta}_1)$ has the ***standard* normal distribution** with mean 0 and variance 1: i.e.,

$$Z(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{Var(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \left(\sum_i x_i^2\right)^{1/2}} \sim N(0,1).$$

<u>Implication 2</u>: The sampling distribution of the OLS intercept coefficient estimator $\hat{\beta}_0$ is *normal* with mean $E(\hat{\beta}_0) = \beta_0$ and variance $Var(\hat{\beta}_0) = \sigma^2 \sum_i X_i^2 / N \sum_i x_i^2$: i.e.,

$$\hat{\beta}_0 \sim N\left(\beta_0, Var(\hat{\beta}_0)\right) = N\left(\beta_0, \frac{\sigma^2 \sum_i X_i^2}{N \sum_i x_i^2}\right).$$

The normality of the sampling distribution of $\hat{\beta}_0$ implies that the **Z-statistic** $Z(\hat{\beta}_0) = (\hat{\beta}_0 - \beta_0)/se(\hat{\beta}_0)$ has the ***standard* normal distribution** with mean 0 and variance 1: i.e.,

$$Z(\hat{\beta}_0) = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{Var(\hat{\beta}_0)}} = \frac{\hat{\beta}_0 - \beta_0}{se(\hat{\beta}_0)} = \frac{\hat{\beta}_0 - \beta_0}{\sigma\left(\sum_i X_i^2\right)^{1/2} / N^{1/2}\left(\sum_i x_i^2\right)^{1/2}} \sim N(0, 1).$$

<u>Implication 3</u>: The statistic $(N-2)\hat{\sigma}^2/\sigma^2$ has a ***chi-square*** **distribution** with (N – 2) degrees of freedom: i.e.,

$$\frac{(N-2)\,\hat{\sigma}^2}{\sigma^2} \sim \chi^2[N-2],$$

where χ^2**[N – 2]** denotes the ***chi-square*** **distribution with (N – 2) degrees of freedom** and $\hat{\sigma}^2$ **is the degrees-of-freedom-adjusted estimator of the error variance** σ^2 given by

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{(N-2)} \qquad \text{where} \quad \hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \quad (i = 1, \ldots, N).$$

<u>Implication 4</u>: The **OLS coefficient estimators** $\hat{\beta}_0$ **and** $\hat{\beta}_1$ **are** ***distributed independently*** **of the error variance estimator** $\hat{\sigma}^2$.

4. Derivation of Test Statistics for $\hat{\beta}_0$ and $\hat{\beta}_1$

4.1 Definition of a Feasible Test Statistic

A **feasible test statistic** must possess **two critical properties**:

(1) It must have a **known distribution**;

(2) It must be capable of being **calculated using only the given sample data** (Y_i, X_i), $i = 1, ..., N$.

We illustrate the derivation of feasible test statistics for $\hat{\beta}_0$ and $\hat{\beta}_1$ by considering the slope coefficient estimator $\hat{\beta}_1$; an analogous argument can be used to obtain a feasible test statistic for $\hat{\beta}_0$.

4.2 A Standard Normal Z-Statistic for $\hat{\beta}_1$

The normality of the sampling distribution of $\hat{\beta}_1$ implies that $\hat{\beta}_1$ can be written in the form of a **standard normal variable**, or **Z-statistic**, with a mean of zero and a variance of one and is denoted as **N(0,1)**.

- **Definition of a Standard Normal Variable**

If some random variable $X \sim N(\mu, \sigma^2)$, then the standardized normal variable defined as $Z = (X - \mu)/\sigma$ has the standard normal distribution N(0,1).

$$X \sim N(\mu, \sigma^2) \quad \Rightarrow \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

- **The standard normal variable, or Z-statistic, for the coefficient estimator $\hat{\beta}_1$**

- The error normality assumption A9 implies that

$$\hat{\beta}_1 \sim N\left(\beta_1, \operatorname{Var}(\hat{\beta}_1)\right) = N\left(\beta_1, \frac{\sigma^2}{\sum_i x_i^2}\right).$$

- The Z-statistic for $\hat{\beta}_1$ is therefore defined as

$$Z(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\operatorname{Var}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\operatorname{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \left(\sum_i x_i^2\right)^{1/2}}$$

where $\operatorname{Var}(\hat{\beta}_1)$ is the ***true* variance of** $\hat{\beta}_1$ and $\operatorname{se}(\hat{\beta}_1) \equiv [\operatorname{Var}(\hat{\beta}_1)]^{1/2}$ is the ***true* standard error of** $\hat{\beta}_1$.

- Since $\hat{\beta}_1$ has the normal distribution, the statistic $Z(\hat{\beta}_1)$ has the standard normal distribution N(0,1): that is,

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_i x_i^2}\right) \quad \Rightarrow \quad Z(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma \big/ \left(\sum_i x_i^2\right)^{1/2}} \sim N(0,1). \qquad (7)$$

- **Problem:** The $Z(\hat{\beta}_1)$ statistic in equation (7) is ***not* a feasible test statistic** for $\hat{\beta}_1$ because it involves the ***unknown* parameter σ**, the square root of the unknown error variance σ^2.

4.3 Derivation of the t-Statistic for $\hat{\beta}_1$

- To obtain a feasible test statistic for $\hat{\beta}_1$, we use the **Student's t-distribution**.

- <u>**General Definition of the t-Distribution**</u>

A random variable with the t-distribution is constructed by dividing

(1) a **standard normal random variable Z**

by

(2) the **square root** of an *independent* **chi-square random variable V** that has been divided by its degrees of freedom *m*

The resulting statistic has the **t-distribution with *m* degrees of freedom**.

Formally:

If (1) $Z \sim N(0,1)$

(2) $V \sim \chi^2[m]$

and (3) Z and V are *independent*,

then the random variable

$$t = \frac{Z}{\sqrt{V/m}} \sim t[m]$$

where t[m] denotes the **t-distribution** (or Student's t-distribution) with ***m*** **degrees of freedom**.

- The ***numerator*** **of a t-statistic** is simply an N(0,1) variable Z.
- The ***denominator*** **of a t-statistic** is the square root of a chi-square distributed random variable divided by its degrees of freedom.

Result:

A **t-statistic** is simply the ratio of a standard normal variable to the square root of an *independent* degrees-of-freedom-adjusted chi-square variable with *m* degrees of freedom.

- **Derivation of the t-Statistic for $\hat{\beta}_1$**

- ***Numerator* of the t-statistic for $\hat{\beta}_1$.** The numerator of the t-statistic for $\hat{\beta}_1$ is the $Z(\hat{\beta}_1)$ statistic (7). It will be convenient to re-write the $Z(\hat{\beta}_1)$ statistic in the form

$$Z(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \left(\sum_i x_i^2\right)^{1/2}} = \frac{\left(\hat{\beta}_1 - \beta_1\right)\left(\sum_i x_i^2\right)^{1/2}}{\sigma} \sim N(0,1) \qquad (8)$$

- ***Denominator* of the t-statistic for $\hat{\beta}_1$.** Implication (3) of the normality assumption implies that the statistic $\hat{\sigma}^2/\sigma^2$ has a degrees-of-freedom-adjusted chi-square distribution with (N – 2) degrees of freedom; that is

$$\frac{(N-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2[N-2] \quad \Rightarrow \quad \frac{\hat{\sigma}^2}{\sigma^2} \sim \frac{\chi^2[N-2]}{(N-2)}. \tag{9}$$

The square root of this statistic is therefore distributed as the square root of a degrees-of-freedom-adjusted chi-square variable with (N – 2) degrees of freedom:

$$\frac{\hat{\sigma}}{\sigma} \sim \left[\frac{\chi^2[N-2]}{(N-2)}\right]^{\frac{1}{2}}. \tag{10}$$

- **The *t-statistic* for $\hat{\beta}_1$.** The t-statistic for $\hat{\beta}_1$ is therefore the ratio of (8) to (10): i.e.,

$$t(\hat{\beta}_1) = \frac{Z(\hat{\beta}_1)}{\hat{\sigma}/\sigma} = \frac{\left(\hat{\beta}_1 - \beta_1\right)\left(\sum_i x_i^2\right)^{1/2}/\sigma}{\hat{\sigma}/\sigma}. \tag{11}$$

The t-statistic for $\hat{\beta}_1$ given by (11) can be rewritten without the unknown parameter σ.

- Since the unknown parameter σ is the divisor of both the numerator and denominator of $t(\hat{\beta}_1)$, multiplication of the numerator and denominator of (11) by σ permits the t-statistic for $\hat{\beta}_1$ to be written as

$$t(\hat{\beta}_1) = \frac{\left(\hat{\beta}_1 - \beta_1\right)\left(\sum_i x_i^2\right)^{1/2}/\sigma}{\hat{\sigma}/\sigma} = \frac{\left(\hat{\beta}_1 - \beta_1\right)\left(\sum_i x_i^2\right)^{1/2}}{\hat{\sigma}}. \tag{12}$$

- Dividing the numerator and denominator of (12) by $\left(\sum_i x_i^2\right)^{1/2}$ yields

$$t(\hat{\beta}_1) = \frac{\left(\hat{\beta}_1 - \beta_1\right)}{\hat{\sigma}/\left(\sum_i x_i^2\right)^{1/2}}. \tag{13}$$

- But the **denominator of (13)** is simply the ***estimated*** **standard error of** $\hat{\beta}_1$; i.e.,

$$\frac{\hat{\sigma}}{\left(\sum_i x_i^2\right)^{1/2}} = \sqrt{\hat{Var}(\hat{\beta}_1)} = \hat{se}(\hat{\beta}_1).$$

- ***Result:*** The **t-statistic for** $\hat{\beta}_1$ takes the form

$$t(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\left(\sum_i x_i^2\right)^{1/2}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\hat{Var}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\hat{se}(\hat{\beta}_1)}. \qquad \textbf{(14)}$$

Note that, unlike the $Z(\hat{\beta}_1)$ statistic in (8), the $t(\hat{\beta}_1)$ statistic in (14) is a ***feasible*** **test statistic** for $\hat{\beta}_1$ because its satisfies both the requirements for a feasible test statistic.

- ***Result:*** The **t-statistic for** $\hat{\beta}_0$ is analogous to that for $\hat{\beta}_1$ and has the same distribution: i.e.,

$$t(\hat{\beta}_0) = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\hat{\operatorname{Var}}(\hat{\beta}_0)}} = \frac{\hat{\beta}_0 - \beta_0}{\hat{\operatorname{se}}(\hat{\beta}_0)} \sim t[N-2]$$

where the estimated standard error for $\hat{\beta}_0$ is

$$\hat{\operatorname{se}}(\hat{\beta}_0) = \sqrt{\hat{\operatorname{Var}}(\hat{\beta}_0)} = \left[\frac{\hat{\sigma}^2 \sum_i X_i^2}{N \sum_i x_i^2}\right]^{\frac{1}{2}}.$$

4.4 Derivation of the F-Statistic for $\hat{\beta}_1$

- A second feasible test statistic for $\hat{\beta}_1$ can be derived from the normality assumption A9 using the **F-distribution**.

- <u>**General Definition of the F-Distribution**</u>

A random variable with the F-distribution is the **ratio of two independent random variables**:

(1) one **chi-square distributed random variable V_1 divided by its degrees of freedom** m_1

and

(2) a second ***independent*** **chi-square distributed random variable** V_2 that also has been **divided by its degrees of freedom** m_2.

The resulting statistic has the **F-distribution with** m_1 ***numerator*** **degrees of freedom and** m_2 ***denominator*** **degrees of freedom**.

Formally:

If (1) $V_1 \sim \chi^2[m_1]$

(2) $V_2 \sim \chi^2[m_2]$

and (3) V_1 and V_2 are *independent*,

then the random variable

$$F = \frac{V_1/m_1}{V_2/m_2} \sim F[m_1, m_2]$$

where $F[m_1, m_2]$ denotes the **F-distribution** (or Fisher's F-distribution) **with *m_1* *numerator* degrees of freedom and *m_2* *denominator* degrees of freedom**.

❑ <u>Derivation</u> of the F-Statistic for $\hat{\beta}_1$

- ***Numerator* of the F-statistic for $\hat{\beta}_1$.** The numerator of the F-statistic for $\hat{\beta}_1$ is the square of the $Z(\hat{\beta}_1)$ statistic (7). Recall that **the *square* of a standard normal N(0,1) random variable has a *chi-square distribution* with *one* degree of freedom**. Re-write the $Z(\hat{\beta}_1)$ statistic as in (8) above:

$$Z(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{\sigma \big/ \left(\sum_i x_i^2\right)^{1/2}} = \frac{\left(\hat{\beta}_1 - \beta_1\right)\left(\sum_i x_i^2\right)^{1/2}}{\sigma} \sim N(0,1). \qquad (8)$$

The *square* of the $Z(\hat{\beta}_1)$ statistic is therefore:

$$\left(Z(\hat{\beta}_1)\right)^2 = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\left(se(\hat{\beta}_1)\right)^2} = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\sigma^2 \big/ \left(\sum_i x_i^2\right)} = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2 \left(\sum_i x_i^2\right)}{\sigma^2} \sim \chi^2[1]. \qquad (15)$$

- ***Denominator* of the F-statistic for $\hat{\beta}_1$.** Implication (3) of the normality assumption implies that the statistic $\hat{\sigma}^2/\sigma^2$ has a degrees-of-freedom-adjusted chi-square distribution with (N – 2) degrees of freedom; that is

$$\frac{(N-2)\,\hat{\sigma}^2}{\sigma^2} \sim \chi^2[N-2] \quad \Rightarrow \quad \frac{\hat{\sigma}^2}{\sigma^2} \sim \frac{\chi^2[N-2]}{(N-2)}. \tag{9}$$

- It is possible to show that the $\chi^2[1]$-distributed statistic $\left(Z(\hat{\beta}_1)\right)^2$ in (15) and the $\chi^2[N-2]$-distributed statistic $(N-2)\hat{\sigma}^2/\sigma^2$ in (9) are **statistically *independent***.

- **The *F-statistic* for $\hat{\beta}_1$.** The F-statistic for $\hat{\beta}_1$ is therefore the ratio of (15) to (9):

$$
\begin{aligned}
F(\hat{\beta}_1) &= \frac{\left(Z(\hat{\beta}_1)\right)^2}{\hat{\sigma}^2/\sigma^2} \\
&= \frac{\left(\hat{\beta}_1 - \beta_1\right)^2\left(\sum_i x_i^2\right)/\sigma^2}{\hat{\sigma}^2/\sigma^2} \\
&= \frac{\left(\hat{\beta}_1 - \beta_1\right)^2\left(\sum_i x_i^2\right)}{\hat{\sigma}^2} \qquad (16) \\
&= \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\hat{\sigma}^2/\sum_i x_i^2} \\
&= \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\hat{Var}(\hat{\beta}_1)} \qquad \text{since } \hat{\sigma}^2/\sum_i x_i^2 = \hat{Var}(\hat{\beta}_1).
\end{aligned}
$$

- ***Result:*** The **F-statistic for** $\hat{\beta}_1$ takes the form

$$F(\hat{\beta}_1) = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\hat{\sigma}^2 \Big/ \left(\sum_i x_i^2\right)} = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\hat{Var}(\hat{\beta}_1)} \sim F[1, N-2]. \qquad \textbf{(17)}$$

Note that, like the $t(\hat{\beta}_1)$ statistic in (14), the $F(\hat{\beta}_1)$ statistic in (17) is a ***feasible*** **test statistic** for $\hat{\beta}_1$ because its satisfies both the requirements for a feasible test statistic.

- **Relationship Between the t-statistic and the F-statistic for $\bar{\hat{\beta}}_j$, j = 0, 1:**

- The ***F-statistic*** **for** $\hat{\beta}_1$ **is the *square* of the *t-statistic*** **for** $\hat{\beta}_1$:

$$F(\hat{\beta}_1) = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\hat{Var}(\hat{\beta}_1)} = \frac{\left(\hat{\beta}_1 - \beta_1\right)^2}{\left(\hat{se}(\hat{\beta}_1)\right)^2} = \left(\frac{\hat{\beta}_1 - \beta_1}{\hat{se}(\hat{\beta}_1)}\right)^2 = \left(t(\hat{\beta}_1)\right)^2.$$

- Similarly, the ***F-statistic*** **for** $\hat{\beta}_0$ **is the *square* of the *t-statistic*** **for** $\hat{\beta}_0$:

$$F(\hat{\beta}_0) = \frac{\left(\hat{\beta}_0 - \beta_0\right)^2}{\hat{Var}(\hat{\beta}_0)} = \frac{\left(\hat{\beta}_0 - \beta_0\right)^2}{\left(\hat{se}(\hat{\beta}_0)\right)^2} = \left(\frac{\hat{\beta}_0 - \beta_0}{\hat{se}(\hat{\beta}_0)}\right)^2 = \left(t(\hat{\beta}_0)\right)^2.$$

- The ***t-distribution*** and the ***F-distribution*** **are also related**.

$$F[1, N-2] = \left(t[N-2]\right)^2 \qquad or \qquad t[N-2] = \sqrt{F[1, N-2]}\,.$$

That is, the F-distribution with 1 numerator degree of freedom and N – 2 denominator degrees of freedom *equals* the square of the t-distribution with N – 2 degrees of freedom. Conversely, the t-distribution with N – 2 degrees of freedom *equals* the square root of the F-distribution with 1 numerator degree of freedom and N – 2 denominator degrees of freedom.

Confidence Interval for β1: Derivation

A **two-step** derivation:

<u>Step 1</u>: Start with a probability statement formulated in terms of $t(\hat{\beta}_1)$, the t-statistic for $\hat{\beta}_1$. This probability statement ***implicitly*** **defines** the two-sided $(1-\alpha)$-level confidence interval for β_1.

<u>Step 2</u>: Re-arrange this probability statement to obtain an equivalent probability statement formulated in terms of β_1 rather than $t(\hat{\beta}_1)$. The resultant probability statement ***explicitly*** **defines** the two-sided $(1-\alpha)$-level confidence interval for β_1.

<u>STEP 1</u>: The **two-sided $(1 - \alpha)$-level confidence interval for β_1** is implicitly defined by the probability statement

$$\Pr\left(-t_{\alpha/2}[N-2] \le t(\hat{\beta}_1) \le t_{\alpha/2}[N-2]\right) = 1 - \alpha \tag{1}$$

where

$1 - \alpha$ = the ***confidence*** **level** attached to the confidence interval;

α = the ***significance*** **level**, where $0 < \alpha < 1$;

$t_{\alpha/2}[N-2]$ = the ***critical value*** of the t-distribution with $(N-2)$ degrees of freedom at the $\alpha/2$ (or $100\alpha/2$ percent) significance level;

and $t(\hat{\beta}_1)$ is the t-statistic for $\hat{\beta}_1$ given by

$$t(\hat{\beta}_1) = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\widehat{\mathrm{se}}(\hat{\beta}_1)}. \tag{2}$$

STEP 2: Express the double inequality inside the brackets in probability statement (4) in terms of β_1 rather than $t(\hat{\beta}_1)$.

$$\Pr\left(-t_{\alpha/2}[N-2] \le t(\hat{\beta}_1) \le t_{\alpha/2}[N-2]\right) = 1-\alpha \qquad \textbf{(3)}$$

(1) Substitute in the double inequality

$$-t_{\alpha/2}[N-2] \le t(\hat{\beta}_1) \le t_{\alpha/2}[N-2]$$

the expression for $t(\hat{\beta}_1)$ given in (2) above:

$$-t_{\alpha/2}[N-2] \le \frac{\hat{\beta}_1 - \beta_1}{\hat{se}(\hat{\beta}_1)} \le t_{\alpha/2}[N-2]. \qquad \text{A}$$

(2) Multiply the double inequality A by the positive number $\hat{se}(\hat{\beta}_1) > 0$:

$$-t_{\alpha/2}\hat{se}(\hat{\beta}_1) \le \hat{\beta}_1 - \beta_1 \le t_{\alpha/2}\hat{se}(\hat{\beta}_1). \qquad \text{B}$$

(3) Subtract $\hat{\beta}_1$ from both sides of inequality (B):

$$-\hat{\beta}_1 - t_{\alpha/2}\hat{se}(\hat{\beta}_1) \le -\beta_1 \le -\hat{\beta}_1 + t_{\alpha/2}\hat{se}(\hat{\beta}_1). \qquad \text{C}$$

(4) Multiply all terms in inequality (C) by -1, remembering to reverse the direction of the inequalities:

$$\hat{\beta}_1 - t_{\alpha/2}\hat{se}(\hat{\beta}_1) \le \beta_1 \le \hat{\beta}_1 + t_{\alpha/2}\hat{se}(\hat{\beta}_1). \qquad \text{D}$$

- Similarly for intercept we have

$$\Pr\left(\hat{\beta}_0 - t_{\alpha/2}[N-2]\hat{se}(\hat{\beta}_0) \le \beta_0 \le \hat{\beta}_0 + t_{\alpha/2}[N-2]\hat{se}(\hat{\beta}_0)\right) = 1 - \alpha .$$

- Basic labor economics theory tells us, that among many variables, education is an important determinant of wages.

Years of Scholling	Mean Hourly wage
6	4.4567
7	5.77
8	5.9787
9	7.3317
10	7.3182
11	6.5844
12	7.8182
13	7.8351
14	11.0223
15	10.6738
16	10.8361
17	13.615
18	13.531

Obs	Y	X	x	y	x_i^2	y_ix_i
1	4.4567	6	−6	−4.218	36	25.308
2	5.77	7	−5	−2.9047	25	14.5235
3	5.9787	8	−4	−2.696	16	10.784
4	7.3317	9	−3	−1.343	9	4.029
5	7.3182	10	−2	−1.3565	4	2.713
6	6.5844	11	−1	−2.0903	1	2.0903
7	7.8182	12	0	−0.8565	0	0
8	7.8351	13	1	−0.8396	1	−0.8396
9	11.0223	14	2	2.3476	4	4.6952
10	10.6738	15	3	1.9991	9	5.9973
11	10.8361	16	4	2.1614	16	8.6456
12	13.615	17	5	4.9403	25	24.7015
13	13.531	18	6	4.8563	36	29.1378
Sum	112.7712	156	0	0	182	131.7856

Obs	X_i^2	Y_i^2	$\hat{Y}_i$	$\hat{u}_i = Y_i - \hat{Y}$	$\hat{u}_i^2$
1	36	19.86217	4.165294	0.291406	0.084917
2	49	33.2929	4.916863	0.853137	0.727843
3	64	35.74485	5.668432	0.310268	0.096266
4	81	53.75382	6.420001	0.911699	0.831195
5	100	53.55605	7.17157	0.14663	0.0215
6	121	43.35432	7.923139	−1.33874	1.792222
7	144	61.12425	8.674708	−0.85651	0.733606
8	169	61.38879	9.426277	−1.59118	2.531844
9	196	121.4911	10.17785	0.844454	0.713103
10	225	113.93	10.92941	−0.25562	0.065339
11	256	117.4211	11.68098	−0.84488	0.713829
12	289	185.3682	12.43255	1.182447	1.398181
13	324	183.088	13.18412	0.346878	0.120324
Sum	2054	1083.376	112.7712	≈0	9.83017

$$x_i = X_i - \bar{X}; y_i = Y_i = \bar{Y}$$

$$\hat{\beta}_2 = \frac{\Sigma y_i x_i}{\Sigma x_i^2} = \frac{131.7856}{182.0} = 0.7240967$$

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X} = 8.674708 - 0.7240967x12 = -0.01445$$

$$\hat{\sigma}^2 = \frac{\Sigma \hat{u}_i^2}{n-2} = \frac{9.83017}{11} = 0.893652; \hat{\sigma} = 0.945332$$

$$\text{var}(\hat{\beta}_2) = \frac{\hat{\sigma}^2}{\Sigma x_i^2} = \frac{0.893652}{182.0} = 0.004910; \text{se}(\hat{\beta}_2) = \sqrt{0.00490} = 0.070072$$

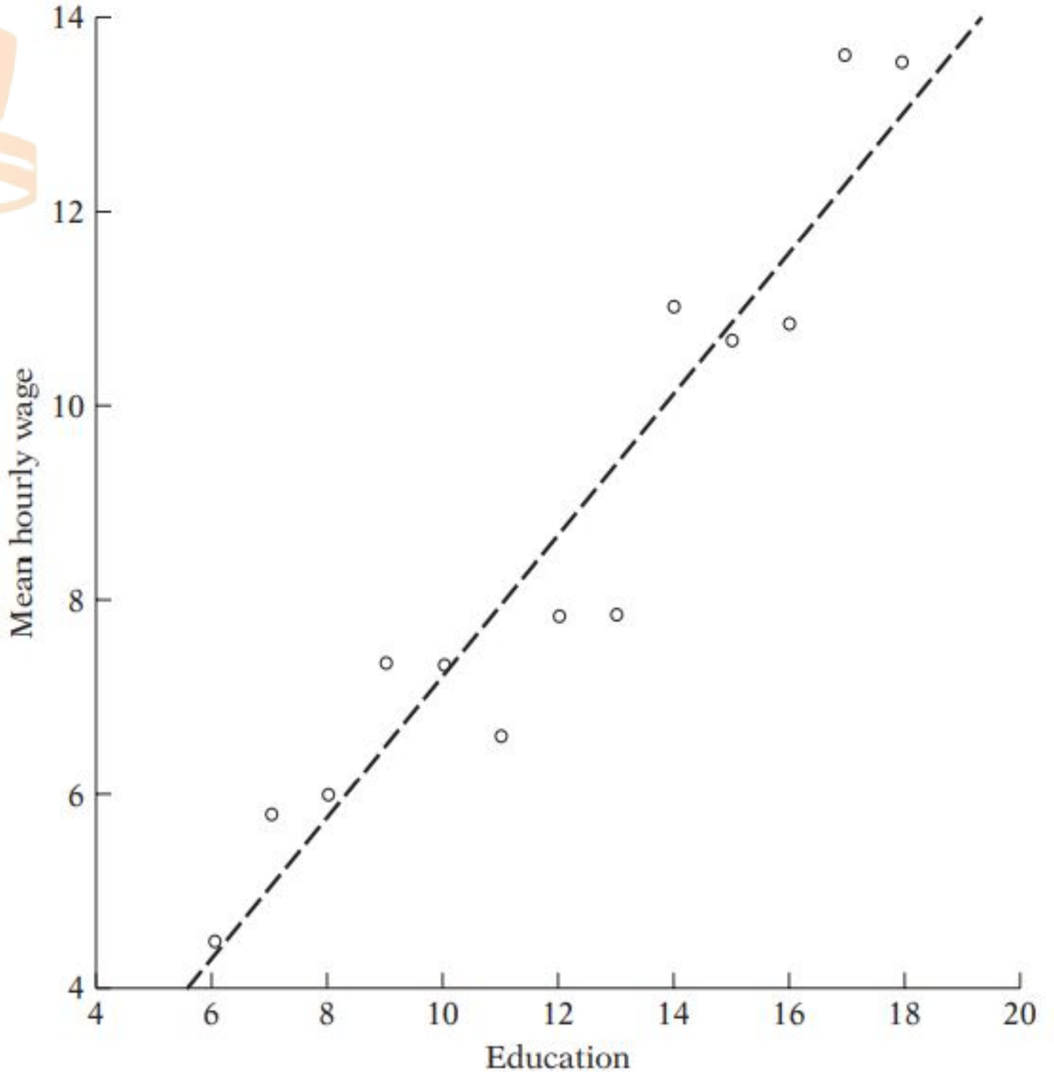
$$r^2 = 1 - \frac{\Sigma \hat{u}_i^2}{\Sigma (Y_i - \bar{Y})^2} = 1 - \frac{9.83017}{105.1188} = 0.9065$$

$$r = \sqrt{r^2} = 0.9521$$

$$\text{var}(\hat{\beta}_1) = \frac{\Sigma x_i^2}{n \Sigma x_i^2} = \frac{2054}{13(182)} = 0.868132;$$

$$\text{se}(\hat{\beta}_1) = \sqrt{0.868132} = 0.9317359$$

FIGURE 3.11
Estimated regression line for wage-education data from Table 2.6.



Returning to our regression example in Chapter 3 (Section 3.6) of mean hourly wages (Y) on education (X), recall that we found in Table 3.2 that $\hat{\beta}_2 = 0.7240$; se $(\hat{\beta}_2) = 0.0700$. Since there are 13 observations, the degrees of freedom (df) are 11. If we assume $\alpha = 5\%$, that is, a 95% confidence coefficient, then the t table shows that for 11 df the **critical** $t_{\alpha/2} = 2.201$. Substituting these values in Eq. (5.3.5), the reader should verify that the 95 percent confidence interval for β_2 is as follows:[4]

$$0.5700 \leq \beta_2 \leq 0.8780 \tag{5.3.9}$$

Or, using Eq. (5.3.6), it is

$$0.7240 \pm 2.201(0.0700)$$

that is,

$$0.7240 \pm 0.1540 \tag{5.3.10}$$

The interpretation of this confidence interval is: Given the confidence coefficient of 95 percent, in 95 out of 100 cases intervals like Equation 5.3.9 will contain the true β_2. But, as warned earlier, we cannot say that the probability is 95 percent that the specific interval in Eq. (5.3.9) contains the true β_2 because this interval is now fixed and no longer random; therefore β_2 either lies in it or it does not: The probability that the specified fixed interval includes the true β_2 is therefore 1 or 0.

Following Eq. (5.3.7), and the data in Table 3.2, the reader can easily verify that the 95 percent confidence interval for β_1 for our example is

$$-1.8871 \leq \beta_1 \leq 1.8583 \tag{5.3.11}$$

Again you should be careful in interpreting this confidence interval. In 95 out of 100 cases, intervals like Equation 5.3.11 will contain the true β_1; the probability that this particular fixed interval includes the true β_1 is either 1 or 0.